

# Esercizio 7.2

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$$H = \omega \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \omega X \quad A = a \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} = a(X - Z) \quad B = b \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = bZ$$

$$C = c \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = c \frac{1+Z}{2}$$

$$[X, Y] = 2iZ$$

$$[Z, X] = 2iY = -[X, Z]$$

A cost del moto  $\Leftrightarrow [H, A] = 0$  ( $\frac{\partial A}{\partial t} = 0$ )

$$\begin{aligned} [H, A] &= HA - AH = \begin{pmatrix} 0 & \omega \\ \omega & 0 \end{pmatrix} \begin{pmatrix} -a & a \\ a & a \end{pmatrix} - \begin{pmatrix} -a & a \\ a & a \end{pmatrix} \begin{pmatrix} 0 & \omega \\ \omega & 0 \end{pmatrix} = \\ &= \begin{pmatrix} a\omega & a\omega \\ -a\omega & a\omega \end{pmatrix} - \begin{pmatrix} a\omega & -a\omega \\ a\omega & a\omega \end{pmatrix} = \begin{pmatrix} 0 & 2a\omega \\ -2a\omega & 0 \end{pmatrix} \neq 0 \end{aligned}$$

$$\begin{aligned} [H, A] &= [\omega X, a(X - Z)] = \omega a [X, X - Z] = \cancel{\omega a [X, X]} - \omega a [X, Z] \\ &= -\omega a (-2iY) = 2\omega a iY = \\ &= 2\omega a i \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = 2\omega a \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \end{aligned}$$

$$[H, B] = [\omega X, bZ] = \omega b [X, Z] = -2i\omega b Y \neq 0$$

$$\begin{aligned} [H, C] &= \begin{pmatrix} 0 & \omega \\ \omega & 0 \end{pmatrix} \begin{pmatrix} c & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} c & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & \omega \\ \omega & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ \omega c & 0 \end{pmatrix} - \begin{pmatrix} 0 & \omega c \\ 0 & 0 \end{pmatrix} = \\ &= \begin{pmatrix} 0 & -\omega c \\ \omega c & 0 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} [H, C] &= \frac{1}{2}\omega c [X, 1+Z] = \frac{1}{2}\omega c [X, Z] \\ &= \frac{1}{2}\omega c (-2iY) = -i\omega c Y \end{aligned}$$

Misura di  $e \rightarrow C |\psi_k^e\rangle = \sum_k |\psi_k^e\rangle \quad \sum_k = 0, e$

$$|\psi_e^e\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle \quad |\psi_0^e\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle = |\psi(0)\rangle \quad \text{stato iniziale}$$

$$H = \omega \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \omega X$$

$$|H - E\mathbb{1}| = 0$$

$\dots + \dots$

$$H = \omega \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \omega X$$

$$|H - E I| = 0$$

$$\begin{vmatrix} -E & \omega \\ \omega & -E \end{vmatrix} = E^2 - \omega^2 = 0 \quad E = \pm \omega$$

$$|\psi_w^H\rangle = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad -x_1 + x_2 = 0 \Rightarrow \begin{cases} x_2 = x_1 = 1/\sqrt{2} \\ |x_1|^2 + |x_2|^2 = 1 \end{cases}$$

$$|\psi_{-w}^H\rangle = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \omega x_1 + \omega x_2 = 0 \Rightarrow \begin{cases} x_1 = -x_2 = 1/\sqrt{2} \\ |x_1|^2 + |x_2|^2 = 1 \end{cases}$$

$$= \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$|\psi_w^H\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$|\psi(0)\rangle = |1\rangle = |\psi_w^H\rangle \langle \psi_w^H | 1 \rangle + |\psi_{-w}^H\rangle \langle \psi_{-w}^H | 1 \rangle$$

$$= \frac{1}{\sqrt{2}} (|\psi_w^H\rangle - |\psi_{-w}^H\rangle) \frac{1}{\sqrt{2}}$$

$$|\psi(t)\rangle = e^{-iHt/\hbar} |\psi(0)\rangle =$$

$$e^{-iHt/\hbar} |\psi_w^H\rangle = e^{-i\omega t/\hbar} |\psi_w^H\rangle$$

$$= e^{-iHt/\hbar} \frac{(|\psi_w^H\rangle - |\psi_{-w}^H\rangle)}{\sqrt{2}} = \frac{1}{\sqrt{2}} \left( e^{-i\omega t/\hbar} |\psi_w^H\rangle - e^{i\omega t/\hbar} |\psi_{-w}^H\rangle \right)$$

$$= \frac{1}{2} \begin{pmatrix} e^{-i\omega t/\hbar} - e^{i\omega t/\hbar} \\ e^{-i\omega t/\hbar} + e^{i\omega t/\hbar} \end{pmatrix} = \begin{pmatrix} -i \sin \omega t/\hbar \\ \cos \omega t/\hbar \end{pmatrix}$$

$$\sinh x = \frac{e^{ix} + e^{-ix}}{2i}$$

$$\cosh x = \frac{e^{ix} + e^{-ix}}{2}$$

$$A = a \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix} \quad |A - \alpha I| = 0$$

$$\begin{vmatrix} -a-\alpha & a \\ a & a-\alpha \end{vmatrix} = -(a+\alpha)(a-\alpha) - a^2 = 0$$

$$\alpha^2 - 2a^2 = 0$$

$$\alpha_{\pm} = \pm \sqrt{2} a$$

$$|\psi_+^A\rangle = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$ax_1 + (a - \sqrt{2}a)x_2 = 0$$

$$x_1 + (1 - \sqrt{2})x_2 = 0$$

$$x_2 = + \frac{1}{-1 + \sqrt{2}} x_1$$

$$x_1 + (1 - \sqrt{2})x_2 = 0 \quad x_2 = + \frac{1}{-1 + \sqrt{2}} x_1$$

$$|\psi_+^A\rangle = \frac{1}{N} \begin{pmatrix} 1 \\ \sqrt{2}-1 \end{pmatrix}$$

$$2(2 - \sqrt{2}) = N^2$$

$$|\psi_-^A\rangle = \frac{1}{\sqrt{2(2+\sqrt{2})}} \begin{pmatrix} 1 \\ \sqrt{2}+1 \end{pmatrix}$$

$$|\psi(t)\rangle = \begin{pmatrix} -i \sin \frac{\omega t}{\hbar} \\ \cos \frac{\omega t}{\hbar} \end{pmatrix}$$

$$P_{+\sqrt{2}\alpha}^A = |\langle \psi_+^A | \psi(t) \rangle|^2 = \frac{1}{N^2} \left| \underbrace{-i \sin \frac{\omega t}{\hbar}} + \underbrace{(\sqrt{2}-1) \cos \frac{\omega t}{\hbar}} \right|^2$$

$$= \frac{1}{2(2-\sqrt{2})} \left( \sin^2 \frac{\omega t}{\hbar} + (2+1-2\sqrt{2}) \cos^2 \frac{\omega t}{\hbar} \right)$$

$$= \frac{1}{2(2-\sqrt{2})} \left[ 1 + 2(1-\sqrt{2}) \cos^2 \frac{\omega t}{\hbar} \right]$$

$$P_{-\sqrt{2}\alpha}^A = \frac{1}{2(2+\sqrt{2})} \left[ 1 + 2(1+\sqrt{2}) \cos^2 \frac{\omega t}{\hbar} \right]$$

$$P_{+\sqrt{2}\alpha}^A + P_{-\sqrt{2}\alpha}^A = 1$$